Fast Eyes, Sharp Eyes, or Good Eyes? What Links Visual Perception to Higher-Order Cognition

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Background

Simple perceptual tasks predict working memory and fluid intelligence. Two abilities have been proposed to explain this:

- **Processing speed:** people differ in how *quickly* they extract visual information. Inspection time (briefly shown, masked targets) correlates ≈ .3 with IQ across 92 studies.
- **Sensory discrimination:** people differ in how *precisely* they resolve a stimulus, with no time limit. Discriminating line lengths or tone pitch correlates .16–.53 with reasoning.

Our approach: *temporal masking* limits processing time, while *spatial masking* limits clarity. The stimuli, judgments and masks are matched, so only the mask type differs.

Method

Participants: 79 UCI undergraduates, tested in three 1-hour sessions on separate days.

6 perceptual tasks = 3 stimuli × 2 mask types. Within each pair, the stimulus, judgment and mask are identical; only the way the mask degrades the target differs.

Stimulus	Judgment	Mask	Spatial difficulty
Pi	which leg is longer?	line segments	leg-length difference
Letter	which of 9 letters?	filtered noise	letter-to-noise weight
Gabor	tilt left or right?	white noise	contrast

- **Temporal masking:** the target is replaced by five masks (48.5 ms each) after a variable SOA; difficulty is the SOA.
- **Spatial masking:** the target is shown embedded in the mask for 485 ms; difficulty is the signal strength.
- **Procedure:** 300 trials per task per person (250 analyzed). Difficulty was adapted trial by trial with two interleaved staircases (targeting 71% and 79% accuracy).
- **4 cognitive tasks:** OSPAN, CSPAN, SSPAN (working memory) and UCMRT (fluid intelligence, matrix reasoning).

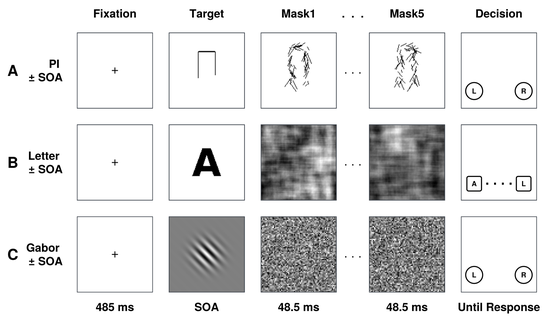


Figure 1. Temporal masking tasks.

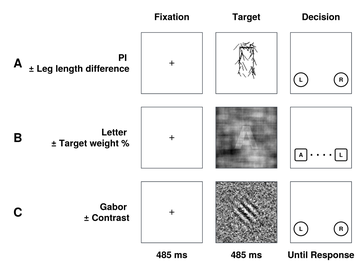


Figure 2. Spatial masking tasks.

Three Hypotheses

Which factor carries the link between perception and cognition?

- **Fast eyes:** temporal masking ability (processing speed). Temporal tasks should cluster and drive the link.
- **Sharp eyes:** spatial masking ability (seeing through noise). Spatial tasks should cluster and drive the link.
- **Good eyes:** a general visual ability shared by all tasks, neither temporal nor spatial.

We expected fast eyes to win.

Hierarchical Psychometric Model

Let i denotes individual, j denotes task, and k denote trials. The model fit to all trial-level data estimates each person's abilities and how they relate, *corrected for trial noise*. It has three levels.

1. Data level. Trial accuracy depends on stimulus strength s and the person's threshold θ on each task:

$$y_{ijk} \sim \text{Bernoulli}\big(\Phi(\beta_j(\theta_{ij} - s_{ijk}))\big)$$

2. Measurement level. The six thresholds θ_i and four cognitive scores c_i form $\gamma_i = (\theta_i', c_i')'$, explained by seven factors η_i (general visual, temporal, spatial, Pi, Letter, Gabor, cognition):

$$\gamma_i = \mu + \Lambda\,\eta_i + \epsilon_i$$

$$\Lambda = \begin{array}{r|ccccccc} & \text{Visual} & \text{Temp.} & \text{Spat.} & \text{Pi} & \text{Letter} & \text{Gabor} & \text{Cog.} \\ \hline \text{Temporal Pi} & \lambda_{11} & \lambda_{12} & 0 & \lambda_{14} & 0 & 0 & 0 \\ \text{Temporal Letter} & \lambda_{21} & \lambda_{22} & 0 & 0 & \lambda_{25} & 0 & 0 \\ \text{Temporal Gabor} & \lambda_{31} & \lambda_{32} & 0 & 0 & 0 & \lambda_{36} & 0 \\ \text{Spatial Pi} & \lambda_{41} & 0 & \lambda_{43} & \lambda_{44} & 0 & 0 & 0 \\ \text{Spatial Letter} & \lambda_{51} & 0 & \lambda_{53} & 0 & \lambda_{55} & 0 & 0 \\ \text{Spatial Gabor} & \lambda_{61} & 0 & \lambda_{63} & 0 & 0 & \lambda_{66} & 0 \\ \text{OSPAN} & 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{77} \\ \text{CSPAN} & 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{87} \\ \text{SSPAN} & 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{97} \\ \text{UCMRT} & 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{10,7} \end{array}$$

Each perceptual task loads on the general factor, one mask factor and one stimulus factor. Cognitive tasks load only on cognition.

3. Structural level. Cognition is regressed on the six perceptual factors (which are independent of each other):

$$\eta_{\text{cog}} = b_1\eta_{\text{general}} + b_2\eta_{\text{temporal}} + b_3\eta_{\text{spatial}} + \cdots + \zeta$$

Large $b_2 \Rightarrow$ fast eyes; large $b_3 \Rightarrow$ sharp eyes; large $b_1 \Rightarrow$ good eyes. Each weight is tested with a Bayes factor.

Result 1: Sources of Individual Differences

Task	General	Masking	Stimulus	Unique
Temporal Pi	.30	.03	.44	.22
Temporal Letter	.17	.25	.39	.19
Temporal Gabor	.29	.22	.20	.30
Spatial Pi	.34	.15	.34	.17
Spatial Letter	.35	.18	.28	.19
Spatial Gabor	.47	.10	.20	.23
Average	**.32**	**.16**	**.31**	**.22**

Proportion of variance by source. The general factor is the largest shared source (32%), then stimulus (31%), and mask type is smallest (16%). Stimulus beats mask with posterior probability .97.

Result 2: Tasks Cluster by Stimulus

Temporal and spatial versions of the same stimulus cluster together first (Pi, then Letter, then Gabor). Neither mask type forms a cluster of its own, and all six perceptual tasks form one group separate from the four cognitive tasks.

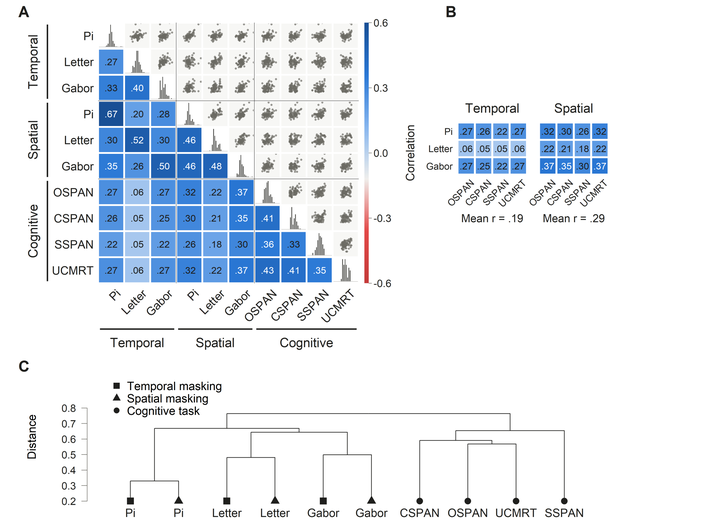


Figure 3. Model-estimated correlations (A, B) and clustering (C).

Spatial tasks correlate more with cognition than temporal tasks (mean r = .29 vs. .19), but this is driven by the temporal Letter task ($r \approx .05$). Without it, temporal r = .25.

Result 3: What Relates to Cognition?

Factor	r [95% CI]	BF_{10}
General visual	**.56 [.15, .84]**	**12.5**
Temporal masking	−.03 [−.44, .39]	0.43
Spatial masking	.19 [−.29, .61]	0.70
Pi	.16 [−.20, .50]	0.57
Letter	−.15 [−.50, .24]	0.53
Gabor	.27 [−.23, .67]	0.95

Only the general visual factor is reliably related to cognition (BF_{10} = 12.5, strong evidence). All five specific factors are inconclusive (BF_{10} = 0.43–0.95).

What Are Good Eyes?

The design cannot say, but candidates include:

- Efficient extraction of the target from noise (both masks act as noise).
- Attention control (Tsukahara et al., 2020).
- Motivation, engagement or test-taking strategy shared by all tasks.

Conclusions

- Neither fast eyes nor sharp eyes is privileged. Temporal masking does not measure processing speed alone: it also requires seeing the target clearly.
- Individual differences are organized by stimulus, but this stimulus-specific variance is unrelated to cognition.
- The link runs through a **general visual factor**: **good eyes**.
- Studies should use **multiple stimuli**, so that stimulus-specific variance averages out.

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